

Ziegler-Nichols PID Tuning

Two classic recipes for tuning PID without a model.

Ogata, *Modern Control Engineering*, Ch. 8.

In this tutorial you will:

- apply the **reaction-curve** (first) method from an open-loop step,
- apply the **ultimate-gain** (second) method from sustained oscillation, and
- compare the two tunings on the closed-loop step response.

Step through with **Ctrl+Enter**, or render a report with **publish**.

Ziegler and Nichols proposed two empirical rules for choosing K_p, T_i, T_d that give roughly a 25% overshoot decay ratio, without needing a full mathematical model of the plant.

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First Method: Reaction Curve (Open-Loop Step Response)

Applicable to plants whose open-loop unit-step response is S-shaped (no integration, no dominant complex poles) – characterized by a **delay time** L and a **time constant** T , read from the point of inflection's tangent line.

Plant: $G(s) = \frac{1}{(s+1)(s+2)(s+3)}$ (no integrator, S-shaped step response).

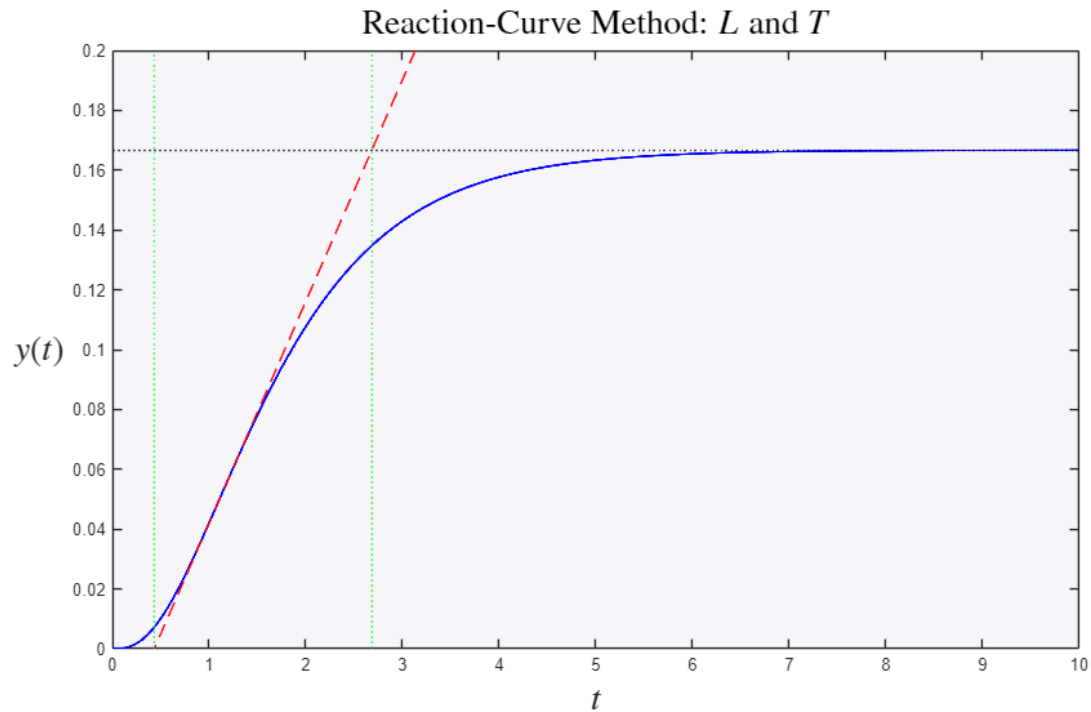
```
G = tf(1,conv(conv([1 1],[1 2]),[1 3]));
[y,t] = step(G,0:0.001:10);

% Find the inflection point (max slope) and its tangent line.
dy = gradient(y,t);
[maxslope,idx] = max(dy);
t_infl = t(idx); y_infl = y(idx);

% Tangent line: y = y_infl + maxslope*(t - t_infl); intersects y=0 at L,
% and reaches the final value at L+T.
yfinal = dcgain(G);
L = t_infl - y_infl/maxslope;
T_rc = (yfinal - y_infl)/maxslope + t_infl - L;
fprintf('Reaction curve: L = %.4f, T = %.4f\n', L, T_rc)

figure
plot(t,y,'b','LineWidth',1.5)
hold on
tang = y_infl + maxslope*(t-t_infl);
plot(t,tang,'r--')
yline(yfinal,'k:')
xline(L,'g:'); xline(L+T_rc,'g:')
hold off
ylim([0 1.2*yfinal])
title('Reaction-Curve Method: $L$ and $T$', 'Interpreter', 'latex', 'FontSize', 20)
ylabel('$y(t)$', 'Interpreter', 'latex', 'FontSize', 20)
set(get(gca, 'YLabel'), 'Rotation', 0)
xlabel('$t$', 'Interpreter', 'latex', 'FontSize', 20)
```

Reaction curve: L = 0.4319, T = 2.2500



Ziegler-Nichols first-method tuning table:

P: $K_p = T/L$

PI: $K_p = 0.9T/L$, $T_i = L/0.3$

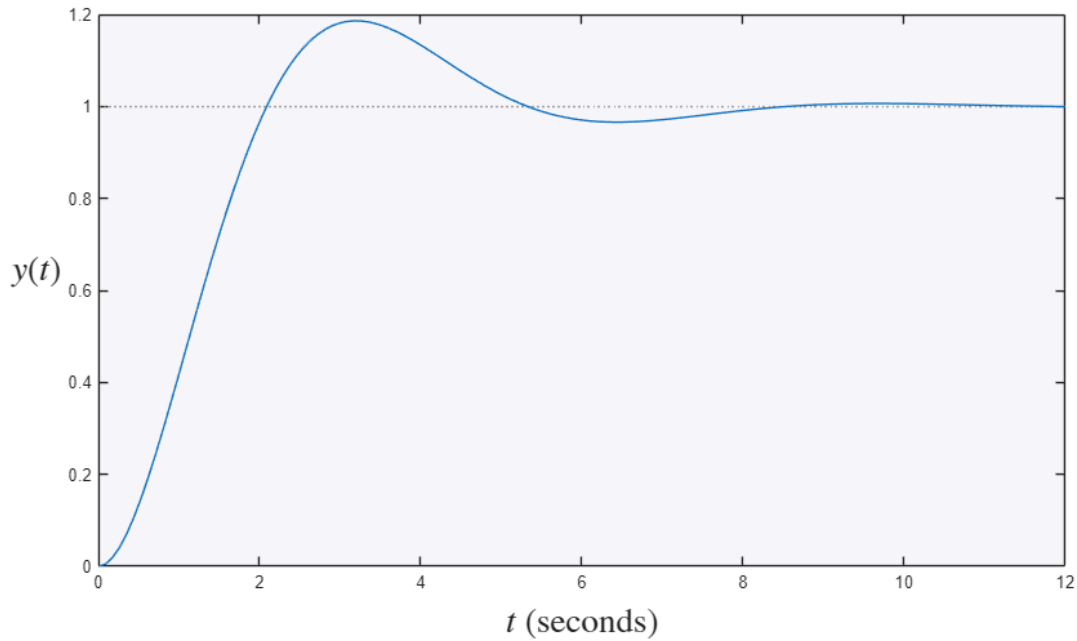
PID: $K_p = 1.2T/L$, $T_i = 2L$, $T_d = 0.5L$

```
Kp_zn1 = 1.2*T_rc/L;
Ti_zn1 = 2*L;
Td_zn1 = 0.5*L;
fprintf('ZN Method 1 PID: Kp=%.4f, Ti=%.4f, Td=%.4f\n', Kp_zn1, Ti_zn1, Td_zn1)

Gc1 = tf(Kp_zn1*[Td_zn1 1 1/Ti_zn1],[1 0]);
T1 = feedback(Gc1*G,1);
figure
step(T1)
title('ZN Method 1: Closed-Loop PID Step Response','Interpreter','latex','FontSize',20)
ylabel('$y(t)$','Interpreter','latex','FontSize',20)
set(get(gca, 'YLabel'), 'Rotation', 0)
xlabel('$t$','Interpreter','latex','FontSize',20)
```

```
ZN Method 1 PID: Kp=6.2508, Ti=0.8639, Td=0.2160
```

ZN Method 1: Closed-Loop PID Step Response



Second Method: Ultimate Gain / Ultimate Period (Closed-Loop)

Applicable more generally: with only proportional control in the loop, raise K_p until the closed-loop response is a sustained oscillation (marginal stability). That gain is K_{cr} , and the oscillation period is P_{cr} .

For $G(s) = \frac{1}{(s+1)(s+2)(s+3)}$, the characteristic equation under proportional gain K is $(s+1)(s+2)(s+3) + K = 0$, i.e. $s^3 + 6s^2 + 11s + (6+K) = 0$. By Routh's array, the marginal-stability gain is found from the s^1 row: $\frac{6 \times 11 - (6+K)}{6} = 0 \Rightarrow K = 60$.

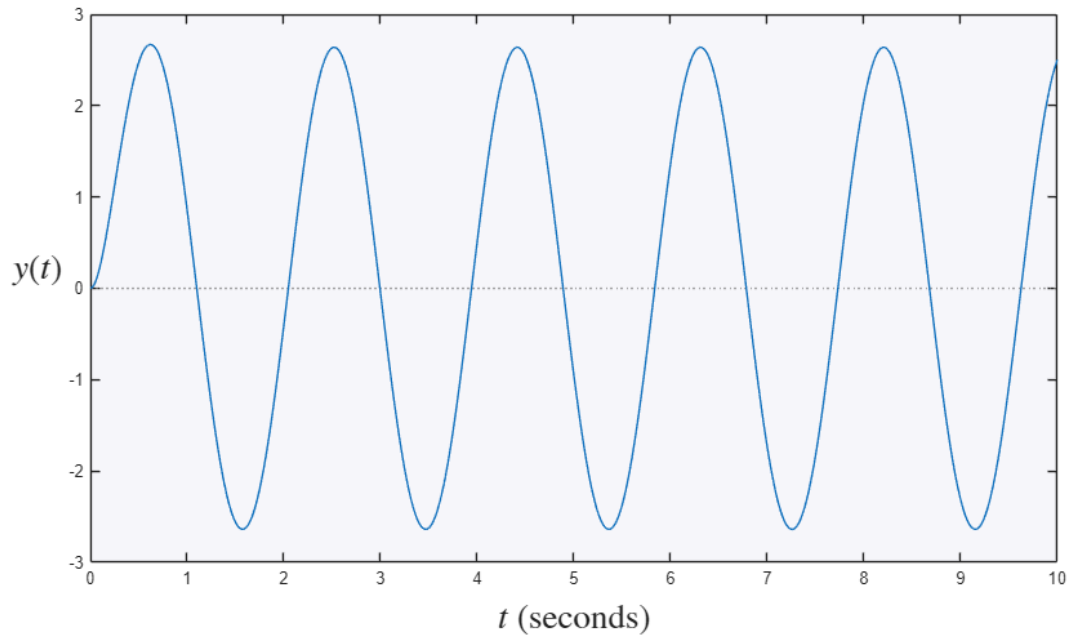
```
Kcr = 60;
fprintf('Ultimate gain Kcr = %.4f\n', Kcr)

% At K=Kcr the auxiliary equation 6s^2+(6+Kcr)=0 gives the oscillation
% frequency.
omega_cr = sqrt((6+Kcr)/6);
Pcr = 2*pi/omega_cr;
fprintf('Ultimate period Pcr = %.4f s (omega_cr = %.4f rad/s)\n', Pcr, omega_cr)

T_marginal = feedback(Kcr*G,1);
figure
impz(T_marginal,0:0.01:10)
title('Sustained Oscillation at $K=K_{cr}$ (Verifying $P_{cr}$)', 'Interpreter','latex', 'FontSize',20)
ylabel('$y(t)$', 'Interpreter','latex', 'FontSize',20)
set(get(gca, 'YLabel'), 'Rotation', 0)
xlabel('$t$', 'Interpreter','latex', 'FontSize',20)

Ultimate gain Kcr = 60.0000
Ultimate period Pcr = 1.8945 s (omega_cr = 3.3166 rad/s)
```

Sustained Oscillation at $K = K_{cr}$ (Verifying P_{cr})



Ziegler-Nichols second-method tuning table:

P: $K_p = 0.5K_{cr}$

PI: $K_p = 0.45K_{cr}$, $T_i = P_{cr}/1.2$

PID: $K_p = 0.6K_{cr}$, $T_i = 0.5P_{cr}$, $T_d = 0.125P_{cr}$

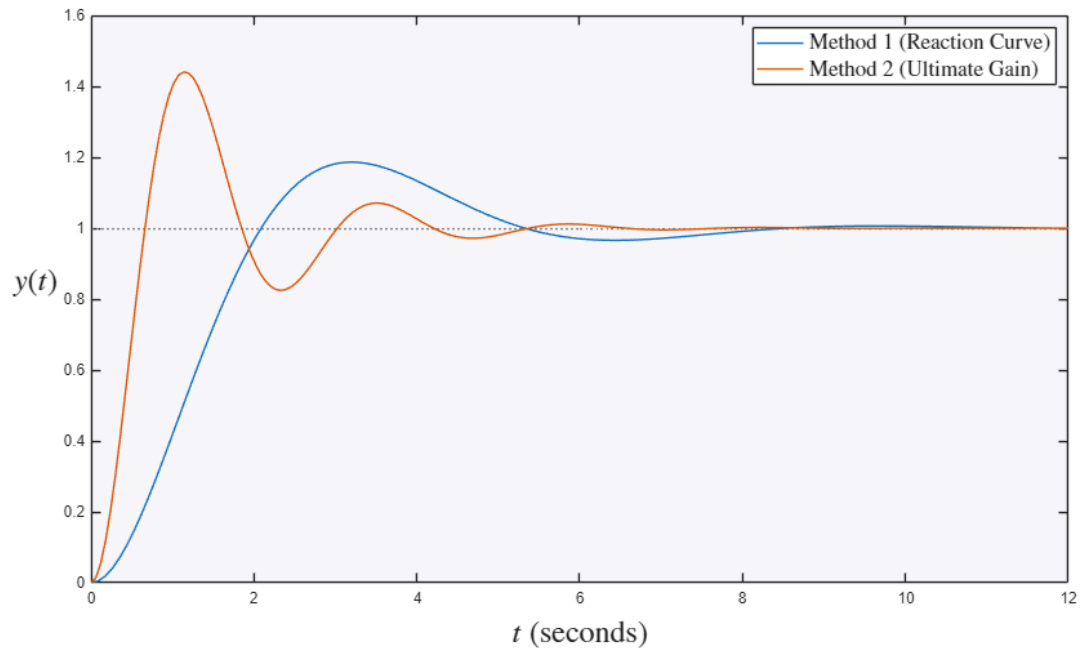
```
Kp_zn2 = 0.6*Kcr;
Ti_zn2 = 0.5*Pcr;
Td_zn2 = 0.125*Pcr;
fprintf('ZN Method 2 PID: Kp=%.4f, Ti=%.4f, Td=%.4f\n', Kp_zn2, Ti_zn2, Td_zn2)

Gc2 = tf(Kp_zn2*[Td_zn2 1 1/Ti_zn2],[1 0]);
T2 = feedback(Gc2*G,1);

figure
hold on
step(T1)
step(T2)
hold off
legend('Method 1 (Reaction Curve)', 'Method 2 (Ultimate Gain)', 'Interpreter', 'latex', 'FontSize', 14)
title('Ziegler-Nichols: Method 1 vs. Method 2 PID Tuning', 'Interpreter', 'latex', 'FontSize', 20)
ylabel('$y(t)$', 'Interpreter', 'latex', 'FontSize', 20)
set(get(gca, 'YLabel'), 'Rotation', 0)
xlabel('$t$', 'Interpreter', 'latex', 'FontSize', 20)
```

```
ZN Method 2 PID: Kp=36.0000, Ti=0.9472, Td=0.2368
```

Ziegler-Nichols: Method 1 vs. Method 2 PID Tuning

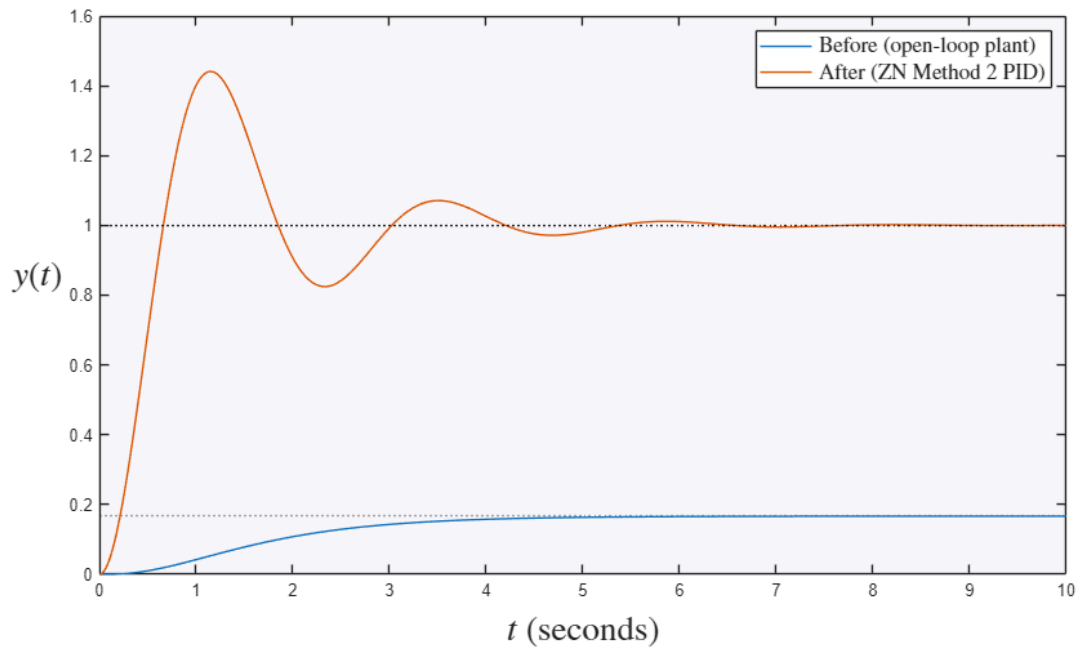


Before vs. After: Uncontrolled Plant vs. ZN-Tuned PID

The payoff: compared with the bare plant's open-loop step (slow, large steady-state error), the Ziegler-Nichols PID tracks the reference with integral action and a reasonable transient.

```
figure
step(G,0:0.01:10)
hold on
step(T2,0:0.01:10)
yline(1,'k:', 'HandleVisibility','off')
hold off
legend('Before (open-loop plant)', 'After (ZN Method 2 PID)', 'Interpreter','latex', 'FontSize',13)
title('Ziegler-Nichols: Before vs. After', 'Interpreter','latex', 'FontSize',17)
ylabel('$y(t)$', 'Interpreter','latex', 'FontSize',20)
set(get(gca, 'YLabel'), 'Rotation', 0)
xlabel('$t$', 'Interpreter','latex', 'FontSize',20)
```

Ziegler-Nichols: Before vs. After



Try it yourself

- Notice both ZN tunings are deliberately aggressive (~25% overshoot); scale K_p down by 0.8 and see a calmer response for a small speed cost.
- Change the plant to add a fourth pole and re-run: the ultimate gain K_{cr} and period P_{cr} shift, and so do the recommended gains.